

Text-Independent Speaker Recognition System

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Abstract—An important step to ensure security of any facility is to grant access only to those who are allowed entrance into the facility and to prevent entrance of others. In this paper, we present a text-independent speaker recognition system. Speaker identification is used to identify the speaker. The model chooses from a collection of a known voice set. There are two types of the application above. The first one is the identification of a closed speaker set. In that case all speakers are known to the system. The second is the identification of an open speaker set that includes previously unknown speakers. Regarding the closed set, a voice from a currently unknown speaker is analyzed by the system and is then compared to the stored speakers. The speaker is classified regarding to the one with the best features. The system compares the voice given at the input in terms of the characteristics of the specific speaker. If the similarity is above a certain threshold then the system accepts the requested identity. In text-independent systems, the system has no prior knowledge of voice phrases that is going to be analyzed. The data during training and verification are different. The system is more flexible but the problem that needs to be solved is harder. In our project, we have collected speeches from 15 students. From those speech samples, we have extracted Mel Frequency Cepstral Coefficient (MFCC) features and other features and trained Patternet Neural Network models. In the testing part, we take speech from a speaker, extract MFCC features and find two minimum distortion matches from the trained Patternet Neural Network models. If the two matches are same, the system identifies the individual. The accuracy of our system is found to be satisfactory for the speakers whom had enough speech samples.

Index Terms—MFCC, MLP, Mel scale, ZCR, Recall, STE

INTRODUCTION

A human's distinctiveness can be identified through various physical characteristics, with voice being a prominent trait. By statistically analyzing a person's recorded voice, numerous features can be derived and utilized for identification purposes. Voice recognition-based security systems have gained popularity owing to their computational efficiency. These systems typically operate by creating a database from speech features extracted from multiple speakers. During testing, a speaker's utterance is processed by the system, which extracts features from the test data and compares them against the database to

determine a match. Feature extraction and matching methods vary significantly based on the nature of speech input. Systems can be text-dependent, requiring speakers to utter specific phrases, or text-independent, not relying on specific words.

The choice of features extracted and the technique used for matching greatly influence the accuracy of speaker identification systems. In this study, we developed a voice recognition-based security system. Such a system is crucial for preventing unauthorized access and managing entry system effectively. Our approach focuses on a text-independent speaker recognition system. We utilized Mel Frequency Cepstral Coefficient (MFCC), Short Time Energy (STE), Zero Crossing Rate (ZCR), Tonalcentroid features for speech feature extraction and employed a Multilayer Perceptron (MLP) neural network-based Pattern Recognition Network (Patternnet) algorithm for data training.

During testing, we utilized Patternnet Training, Trainrp, and Crossentropy for analysis. In the previous version, we employed KNN (The k-nearest neighbors - KNN algorithm is a non-parametric, supervised learning classifier, which uses proximity to make classifications or predictions about the grouping of an individual data point) to detect user. But the results showing for testing purpose was not satisfactory. So, we moved to neural network for better accuracy.

BACKGROUND RESEARCH

Speaker recognition have been a hot topic for quite a long time and several methods have been developed to implement this. In all the methods, the human vocal tract is modeled as a system and features of that system have been extracted. Some voice features are LPC (Linear Predictive Coding), MFCC (Mel Frequency Cepstral Coefficients), SSC (Spectral Sub-band Centroid), Probabilistic Linear Discriminate Analysis (PLDA), Discrete Wavelet Transform (DWT) etc. Various algorithms and classifiers have been used to match the extracted features. Some of them are Hidden Markov Model (HMM), Gaussian Mixture Model (GMM), Vector Quantization (VQ), Dynamic Time Warping (DTW), LGB classifier, KNN classifier, Artificial Neural Networks (ANN) etc. Several Research

have been done to find the best combination of feature and feature matching algorithms. These researches have yielded various levels of accuracy. Some of these well-performing combinations are LPCC and LGB classifier [1], LPCC, SSC and GMM classifier [2], SVM/HMM based speaker recognition [3] etc. In this paper, we have used MFCC features and Neural Network based Patternnet.

PROPOSED METHODOLOGY

A. Data collection and Data cleaning

The first and most crucial stage involved gathering speech samples. We gathered data from 20 pupils roughly speech samples. After sample collecting, we had to remove any subpar samples. These low-quality samples were either very noisy, had a very low loudness, or only partially captured the spoken word or didn't have enough duration of speech. After all these processes, we trimmed down to speech samples of 15 pupils. After that, we divided the data into two portions. One portion for training and the other portion for testing for false positives and true negatives.

B. Voice Activity Detection

Voice Activity Detection (VAD) is a technique used in speech processing that detects the presence or absence of human speech. The main uses of VAD are in speech coding and speech recognition. It can facilitate speech processing and can also be used to turn off certain processes during non-speech in an audio session: it can avoid unnecessary encoding / transmission of silent packets, saving network usage and bandwidth. In this application VAD was implemented using the parameters STE (short time energy) and ZCR (Zero Crossing Rate). The speech signal consists of voice and silence areas. Furthermore, the energy associated with the vocal area is large compared to the silent area, which will have minimal or negligible energy. Thus, short-term energy can be used for speech, noise and silence.

$$E = \sum_{m=-\infty}^{\infty} x^2(m)$$

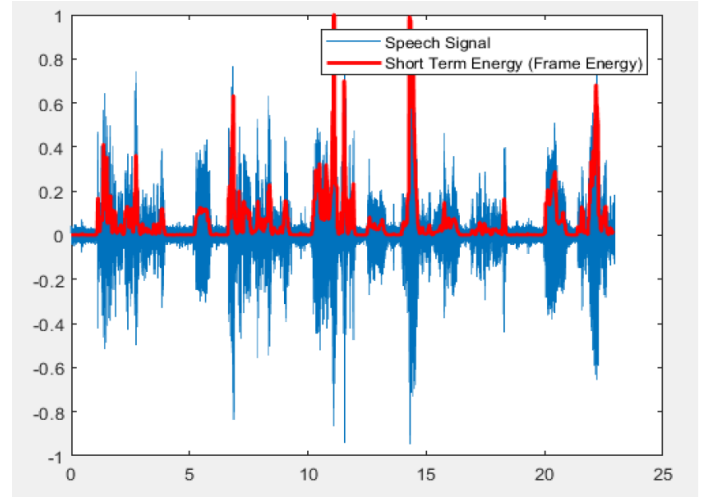
$$E_n = \sum_{m=n-N+1}^n x^2(m) = x^2(n-N+1) + \dots + x^2(n)$$

In the expression above E represents energy of the signal $x(m)$. There is little or no effectiveness of this definition for time-varying signals, such as speech. Therefore we divide our signal into small "windows" whose energy is calculated as follows:

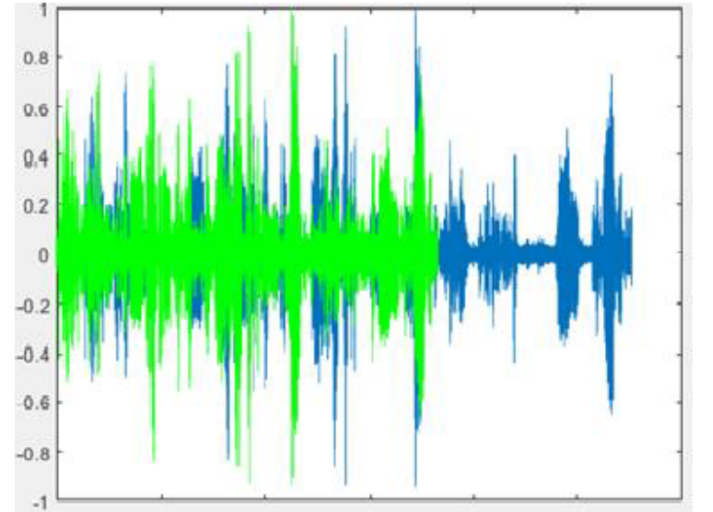
$$E_n = \sum_{m=n-N+1}^n [x(m)w(n-m)]^2$$

Where $w(n-m)$ is the time window, n is the sample in which the analysis window is centered, and N is the window length.

High energy will be classified as vocal while low energy will be classified as non-vocal.



In conclusion, the combination of these 2 parameters with the use of the correct thresholds resulted in the removal of silence from the signals and kept only the parts to which the respective speaker was speaking.

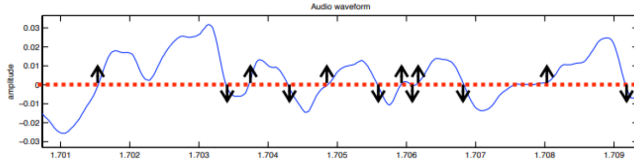


C. Feature Extractions

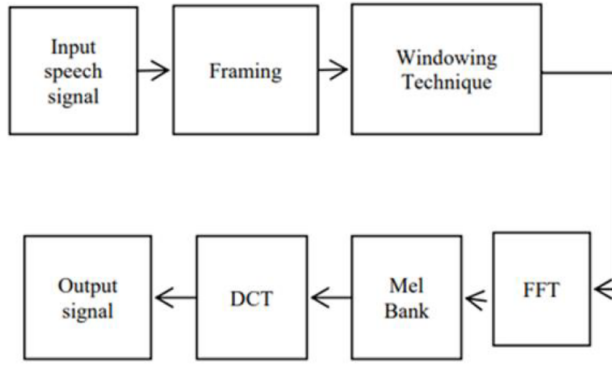
A key point in speech recognition is the features, extracted from the audio signal. The part of feature extraction is very important as it essentially converts our audio signal into parameters, which it represents and which we can then use to further analyze and process our data. A brief description of the features and how they were used will follow, if necessary.

1) *STE: Short Time Energy*: The energy of the signal in a given window.

2) *Zero Crossing Rate*: Zero Crossing Rate: The zero-crossing rate is the rate of sign-changes along a signal, i.e., the rate at which the signal changes from positive to zero to negative or from negative to zero to positive, in a given time-window.



3) **MFCC**: Mel Frequency Cepstral Coefficient (MFCC): The MFCC feature extraction technique is relatively easy to apply, not complicated at all and gives very good results, which is why it is the most common one. It is based on the spectral content of the voice signal, as it is recognized by human hearing. According to psychoacoustic research the human perception of the frequencies of a voice signal does not follow a linear scale. Thus, for every tone with real frequency f measured in Hz, there is a subjective tone on the mel scale, which will be analyzed below. The following figure shows the steps to extract the MFCCs.



1st Step: Initially the method passes the audio signal through a filter that emphasizes high frequencies. Therefore the energy at these frequencies is increased. The higher frequencies in a speech sample are usually low in power. So, pre-emphasis is done to increase the energy of higher frequencies. The filter equation of pre-emphasis is:

$$y[n] = x[n] - 0.99x[n]$$

Here, $x[n]$ is the speech signal, and $y[n]$ is the filtered signal.

2nd Step: For proper Fourier transform, a Hamming window is applied to each frame. Then the power spectral density is calculated. The expression for the Fourier transform is:

$$S(k) = \sum_{n=0}^{N-1} s(n) \cdot w(n) \cdot e^{-j2\pi k/N}$$

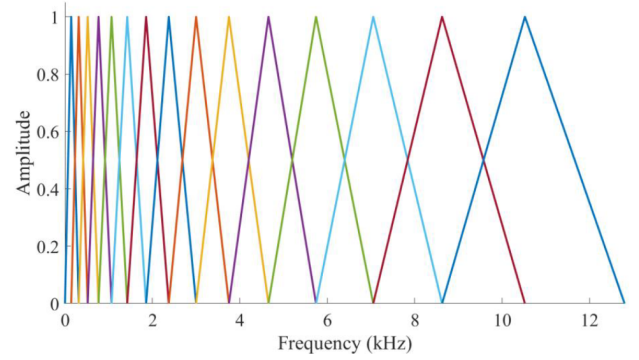
Here, $s(n)$ is the speech frame, $w(n)$ is the Hamming window, and N is the length of the frame. The expression for power spectral Density is:

$$P(k) = \frac{1}{N} |S(k)|^2$$

3rd Step: Fast Fourier Transform is applied to each of these windows and is multiplied by triangular-band filters. Specifically, we apply triangular filters whose center frequencies

distance is equal for the low frequencies and logarithmically placed at the higher frequencies. This is the Mel scale, which defines linear distances below 1000Hz, and logarithmic distances above 1000Hz. The image below shows the filter layout on this scale. The number of mel bandwidth filters selected was 40. This process will also reduce the features involved. The formula that gives the frequency on the Mel scale is the following:

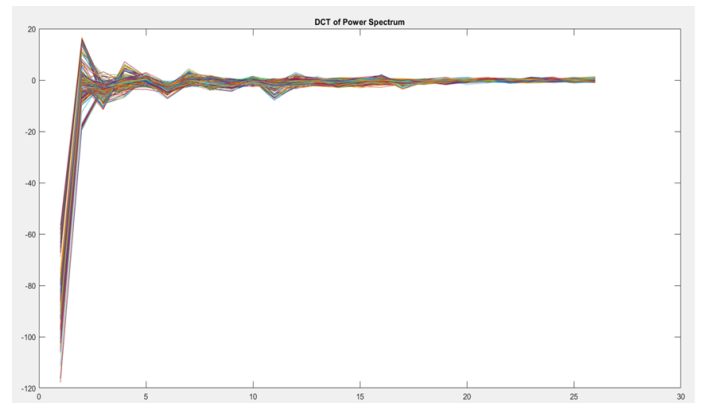
$$Mel(f) = 2595 \log\left(1 + \frac{f}{700}\right)$$



4th step:

The final step in extracting the features is the calculation of MelCepstrum. Specifically, we calculate the natural logarithm of the mel spectrum, and then return it to the time domain using the Discrete Cosine Transform. The advantage of this is that the MFC coefficients will not be related to each other. The representation of the spectral content of a signal through the cepstrum gives a very good description of the spectral properties of the signal in a time-limited analysis window.

$$C_n = E_k \cos\left(\sum_{n=0}^N \left(n \left(k - \frac{1}{2}\right) \frac{\pi}{40}\right)\right); \quad n = 0, 1, 2, \dots, N \quad (1)$$



In order to take into account the changes in the envelope of the audio signals (or the change of the MFCC coefficients) 39

total MFCC characteristics were used in our application, as we see in the table below.

Characteristics Number of them Delta MFCCs indicate the

TABLE I
MFCC, DELTA-MFCC, AND DELTA-DELTA-MFCC

Feature	Number of Coefficients
MFCC	13
Delta-MFCC	13
Delta-delta-MFCC	13
Total	39

speed of change while the delta deltas indicate the acceleration. A significant vocal characteristic is the variation of energy in a frequency range over time, which is reflected in the changes of MFCC coefficients between consecutive frames. These changes are represented by delta coefficients, and the way they evolve over time is captured by delta-delta coefficients. To calculate delta coefficients, the difference between consecutive MFCC coefficients is computed and smoothed using a triangular window. The resulting values indicate the rate of change of spectral features and can provide insights into dynamic aspects of speech like pitch and intonation. However, they are also sensitive to noise and other sources of variability and require careful interpretation and use. The expression for delta coefficients is:

$$d = \frac{\sum_{n=1}^N n(C_{t+n} - C_{t-n})}{2 \sum_{n=1}^N n^2}$$

Here, d represents the delta coefficient of frame t ; C_{t+n} is the MFCC coefficient of frame $t + n$. n is the number of frames included to calculate the delta. From 26 MFCC coefficients, we get 26 delta and 26 delta-deltas. However, only 12 of the 26 MFCCs are necessary. So, we are left with 12+12+12=36 coefficients from a single frame. If the frame number is F , then we get a matrix of dimension 36 x F from a single speech sample.

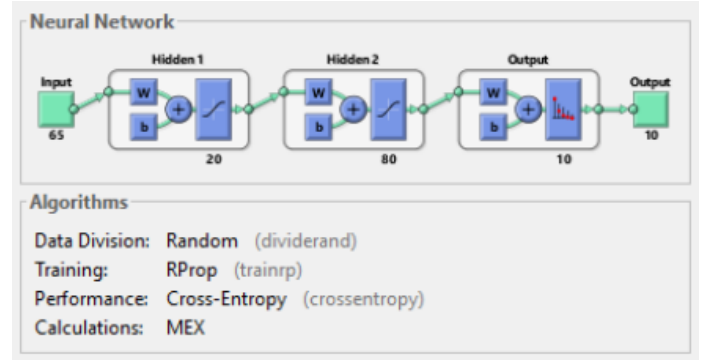
D. Neural Network Implementation

In regards of the Neural Network, two source code files have been developed. The first file implements the search for the optimal number of features, with which the model will be trained, while the second concerns the implementation of the final model. Both files concern the creation of neural networks that get trained with the help of **Patternet** function. Specifically, **Multi-hidden Layers Perceptrons** are used (MLPs). This means that the number of neurons per hidden layer must be selected. After experiments, **20.80** was found to be the ideal number of neurons for first and second hidden layer respectively. The input of the network consists of nodes equal in number each time with the features selected, while the output consists of ten nodes one for each speaker in the data set. The arguments of the patternet function are described below:

Hidden Level Neurons (hiddenLayerSize): the table [20 80].

Training Function (trainFcn): the network training method, ie the way the weights between the neurons are updated at each iteration of the training process. ‘Trainrp’ is used, ie a back propagation method. The update of the weights is done with by minimizing a cost function (here the difference between actual class and prediction).

Performance Function (performFcn): a metric of network quality, using crossentropy. In information theory it counts the difference between two probability distributions. In the classification models, each record belongs with probability 1 in one class and 0 in the others. Each model estimates the probability that a record belongs to a class. Cross-entropy is the difference between the two distributions. Its minimization is identical to the maximization of the likelihood function.



RESULTS AND DISCUSSION

In this section, we present the results of our experiments and discuss their implications.

E. Experimental Result

For each frame in our project, we extracted 39 MFCC coefficients as features. These coefficients were then stored in a database for further processing.

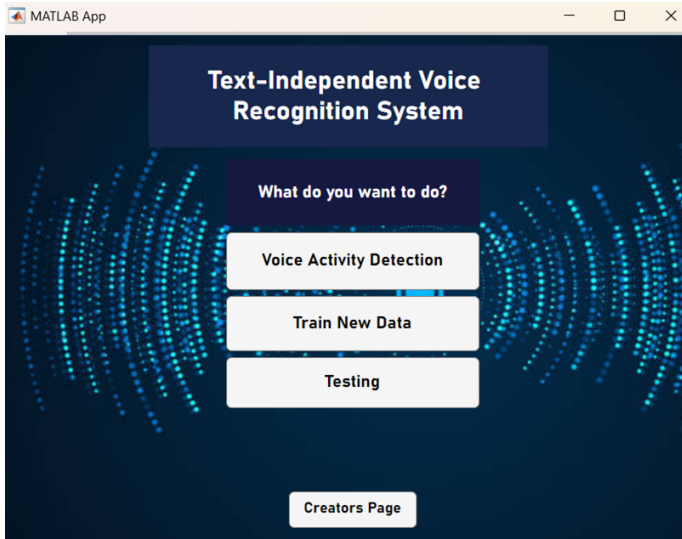
We developed a pattern recognition model using the PatternNet architecture and trained it using the Resilient Back-propagation (trainrp) algorithm. The training data consisted of samples from enlisted individuals, while the testing data included both enlisted and non-enlisted samples, referred to as "true data" and "false data," respectively.

During the testing phase, access was granted if the model determined that the provided ID and Name matched the stored data, and access was denied otherwise.

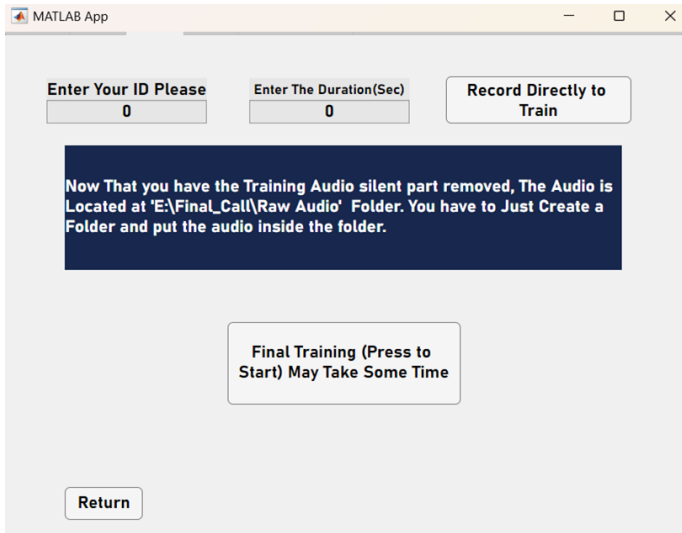
The performance of the model was evaluated based on metrics such as accuracy, precision, and recall, which were calculated using a separate validation dataset.

F. Workspace of Our Model(GUI)

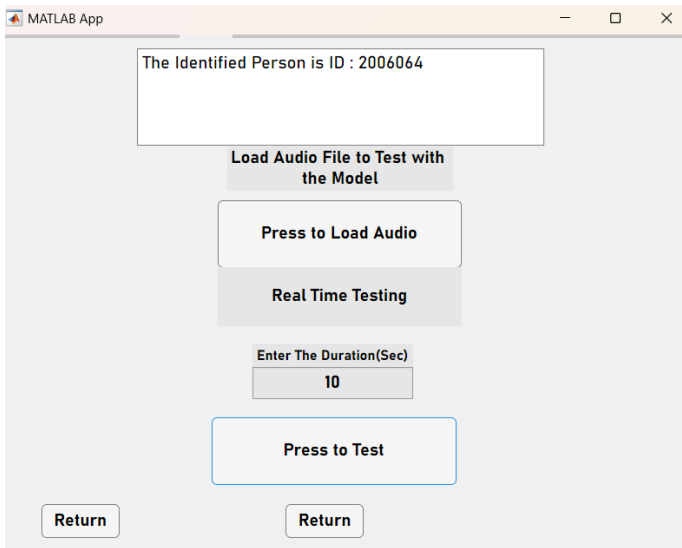
To implement the system easily we developed a Graphical User Interface (GUI) that provides the scope to train and test new dataset easily both in real time.



Homepage



Training Page



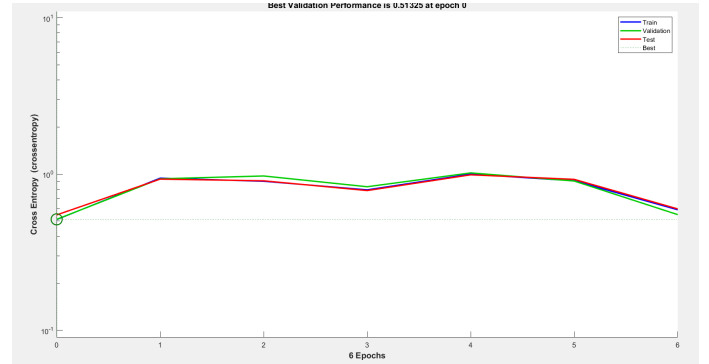
Testing Page

G. Dataset Length

TABLE II
TRAINING AND TESTING SET INFORMATION

Category	Training Set	Testing Set
Number of Speakers	15 Speakers	15 Speakers
Sampling frequency	44.1kHz	44.1kHz
Total duration	8 minutes 40"	3 minutes 2"
Duration per Speaker	~34"	~13"
Total Size	16MB	5.8MB

H. Training Validation Confusion Chart



Training Validation Graph

The Confusion Matrix is created with arguments to the actual class for each record and the predicted. For each speaker, it is found which class was predicted the most times. The performance of the model at class level is 97.53 percent . Then the Accuracy for each speaker is calculated, as well as the average Accuracy that reflects the models quality. This is at 83.99 percent.

Precision, $P = \frac{TP}{(TP+FP)} = \frac{272}{(272+9)} = 96.80$ Here, TP: True positive means predicted positive, actually positive FP: False Positive means predicted positive, actually negative

FUTURE SCOPE

While the current implementation of the text-independent speaker recognition system using MFCC and PatternNet neural network model has shown satisfactory results, there are several avenues for future exploration and improvement:

- 1) **Enhanced Feature Extraction Techniques:** Investigate advanced feature extraction techniques beyond MFCC, such as PLP (Perceptual Linear Prediction) or Gammatone filterbanks, to capture more nuanced characteristics of speech signals.
- 2) **Model Optimization:** Explore different neural network architectures and optimization algorithms to improve the accuracy and efficiency of the PatternNet model. Techniques such as dropout regularization, batch normalization, and hyperparameter tuning could be employed to enhance model performance.
- 3) **Robustness to Environmental Variability:** Enhance the system's robustness to environmental variability, such as background noise, channel distortion, and speaker

position, by incorporating techniques such as data augmentation, multi-condition training, and environmental compensation methods.

- 4) **Integration with Multi-Modal Systems:** Investigate the integration of the speaker recognition system with other biometric modalities, such as facial recognition or fingerprint recognition, to create a more robust and secure multi-modal authentication system.

By exploring these areas of future research and development, the text-independent speaker recognition system can be further refined and enhanced to meet the evolving needs of various applications and industries.

CONCLUSION

In conclusion, the development of a text-independent speaker recognition system using MFCC and PatternNet neural network model has shown promising results in identifying speakers based on their voice characteristics. However, the project encountered several challenges that warrant further attention and exploration. A minimum of seven sets of data from each of the 15 individuals in the audio collection were gathered. Test audio data was utilized for verification after applying the feature extraction method using MFCC. The implementation of the PatternNet neural network model facilitated the verification process. One of the primary challenges faced during the project was the variability in speech lengths within the dataset. This variability posed difficulties in standardizing the input data for model training and testing, leading to inconsistencies in performance across different speech samples. Additionally, users with shorter speech samples presented a specific challenge, as the models struggled to effectively learn and generalize from limited data. Despite these challenges, the project has provided valuable insights into the complexities of speaker recognition technology and has laid the groundwork for future research and development.

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